Car Database Management System

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Abstract

This project focuses on designing, securing, optimizing, and automating the management of an Information System for a used car dataset. It includes preparing a relational schema, implementing security measures, optimizing queries, and automating tasks. Furthermore, the dataset serves as a basis for machine learning applications, including price prediction with linear regression.

Chapter 1

Introduction

Managing information systems for used cars can be complex due to the diverse nature of the data and the need for security, optimization, and automation. This project organizes the data using a relational schema, enforces security policies, optimizes queries for efficient data retrieval, and provides a graphical interface for end-user interaction.

1.1 Dataset Details

The dataset used in this project is sourced from Kaggle. It includes the following columns:

- `name`
- `year`
- `selling_price`
- `km_driven`
- `fuel`
- `seller_type`
- `transmission`
- `Owner`

The dataset is provided as five CSV files.

Chapter 2

Relational Schema

The relational schema for the database is designed to store information across five primary tables:

2.1 Entities

1. **Entities:**
   - Cars
   - Features
   - Specifications
   - Dimensions
   - Ownership

2.2 Tables

1. **Cars Table:**
   - CAR_ID (Primary Key)
   - MAKE
   - MODEL
   - PRICE
   - YEAR
   - FUEL_TYPE
   - LOCATION
   - TRANSMISSION
2. **Features Table:**
   - FEATURE_ID (Primary Key)
   - CAR_ID (Foreign Key)
   - COLOR

- SEATING_CAPACITY
- FUEL_TANK_CAPACITY

3. **Specifications Table:**
    - SPEC_ID (Primary Key)
    - CAR_ID (Foreign Key)
    - ENGINE
    - MAX_POWER
    - MAX_TORQUE
    - DRIVETRAIN

4. **Dimensions Table:**
    - DIMENSION_ID (Primary Key)
    - CAR_ID (Foreign Key)
    - LENGTH
    - WIDTH
    - HEIGHT

5. **Ownership Table:**
    - OWNERSHIP_ID (Primary Key)
    - CAR_ID (Foreign Key)
    - OWNER
    - SELLER_TYPE
    - KILOMETER

2.3 Constraints:

Table 2.1: Constraints

Constraint Name	**Constraint Type**	**Search Condition**
CHK_PRICE	Check	Price ≥ 0
CHK_YEAR	Check	Year BETWEEN 1886 AND 9999
SYS_C008363	Primary Key	(null)
SYS_C008364	Check	CAR_ID IS NOT NULL
SYS_C008365	Check	MAKE IS NOT NULL
SYS_C008366	Check	MODEL IS NOT NULL
SYS_C008367	Check	PRICE IS NOT NULL
SYS_C008368	Check	YEAR IS NOT NULL
SYS_C008369	Check	FUEL_TYPE IS NOT NULL
SYS_C008370	Check	LOCATION IS NOT NULL
SYS_C008371	Check	TRANSMISSION IS NOT NULL

Chapter 3

Security and User Management

3.1 Overview

This section details user and role management in the car management system, ensuring appropriate access levels to database tables for different users. The system accommodates three user types: admin, manager, and regular users, each with distinct roles and permissions.

3.2 Users and Roles

1. **Admin User (C##admin_user)**
   - **Role:** C##admin_role
   - **Permissions:** Full access to all tables (CarDetails, CarSpecifications, CarFeatures, CarOwnership, CarDimensions).
2. **Manager User (C##manager_user)**
   - **Role:** C##manager_role
   - **Permissions:** Modify access (read, insert, update) to all tables, with no Data Definition Language (DDL) permissions.
3. **Regular User (C##regular_user)**
   - **Role:** C##regular_role
   - **Permissions:** Read-only access to all tables.

3.3 Role Creation and Privileges

- **Admin Role (C##admin_role):** Full access to all tables.
- **Manager Role (C##manager_role):** Select, insert, and update privileges on all tables.
- **Regular Role (C##regular_role):** Select privileges on all tables.

3.4 Assigning Roles to Users

- C##admin_role is assigned to C##admin_user.
- C##manager_role is assigned to C##manager_user.
- C##regular_role is assigned to C##regular_user.

3.5 Password Policy

The default profile is modified to set a 90-day password lifetime, requiring periodic password changes.

3.6 Views for Restricted Access

A view, `CarDetails_View`, is created to limit access to specific columns (Car_ID, Make, Model) for users with the C##regular_role, ensuring the underlying table remains secure by restricting access to other columns.

Chapter 4

Queries and Optimization

This section demonstrates SQL queries for data retrieval and optimization, including car listings, average price calculations, and query performance improvements. Optimized queries are documented in the project repository here.

	GRANTEE	OWNER	TABLE_NAME	GRANTOR	PRIVILEGE	GRANTABLE	HIERARCHY	COMMON	TYPE	INHERITED
1	C##MANAGER_ROLE	C##CAR_SCHEMA	CARDETAILS	C##CAR_SCHEMA	INSERT	NO	NO	NO	TABLE	NO
2	C##MANAGER_ROLE	C##CAR_SCHEMA	CARSPECIFICATIONS	C##CAR_SCHEMA	INSERT	NO	NO	NO	TABLE	NO
3	C##MANAGER_ROLE	C##CAR_SCHEMA	CARFEATURES	C##CAR_SCHEMA	INSERT	NO	NO	NO	TABLE	NO
4	C##MANAGER_ROLE	C##CAR_SCHEMA	CAROWNERSHIP	C##CAR_SCHEMA	INSERT	NO	NO	NO	TABLE	NO
5	C##MANAGER_ROLE	C##CAR_SCHEMA	CARDIMENSIONS	C##CAR_SCHEMA	INSERT	NO	NO	NO	TABLE	NO
6	C##MANAGER_ROLE	C##CAR_SCHEMA	CARDETAILS	C##CAR_SCHEMA	SELECT	NO	NO	NO	TABLE	NO
7	C##MANAGER_ROLE	C##CAR_SCHEMA	CARSPECIFICATIONS	C##CAR_SCHEMA	SELECT	NO	NO	NO	TABLE	NO
8	C##MANAGER_ROLE	C##CAR_SCHEMA	CARFEATURES	C##CAR_SCHEMA	SELECT	NO	NO	NO	TABLE	NO
9	C##MANAGER_ROLE	C##CAR_SCHEMA	CAROWNERSHIP	C##CAR_SCHEMA	SELECT	NO	NO	NO	TABLE	NO
10	C##MANAGER_ROLE	C##CAR_SCHEMA	CARDIMENSIONS	C##CAR_SCHEMA	SELECT	NO	NO	NO	TABLE	NO
11	C##MANAGER_ROLE	C##CAR_SCHEMA	CARDETAILS	C##CAR_SCHEMA	UPDATE	NO	NO	NO	TABLE	NO
12	C##MANAGER_ROLE	C##CAR_SCHEMA	CARSPECIFICATIONS	C##CAR_SCHEMA	UPDATE	NO	NO	NO	TABLE	NO
13	C##MANAGER_ROLE	C##CAR_SCHEMA	CARFEATURES	C##CAR_SCHEMA	UPDATE	NO	NO	NO	TABLE	NO
14	C##MANAGER_ROLE	C##CAR_SCHEMA	CAROWNERSHIP	C##CAR_SCHEMA	UPDATE	NO	NO	NO	TABLE	NO
15	C##MANAGER_ROLE	C##CAR_SCHEMA	CARDIMENSIONS	C##CAR_SCHEMA	UPDATE	NO	NO	NO	TABLE	NO

Figure 4.1: Optimization of query

Chapter 5

Automation

5.1 Overview

Automation in this project includes triggers, functions, and procedures for managing and streamlining database operations. Examples include:

- Logging deletions.
- Restricting unauthorised actions.
- Calculating total car values.
- Counting cars based on criteria.

Chapter 6

Graphical Interface

6.1 Overview

Two GUI applications, `gui_system.py` and `Bonus.py`, were developed using Python's Tkinter and cx_Oracle libraries for dynamic car data management. Features include:

- Dynamic filtering with dropdown menus.
- Real-time query execution.
- CSV export functionality.

6.2 Screenshots

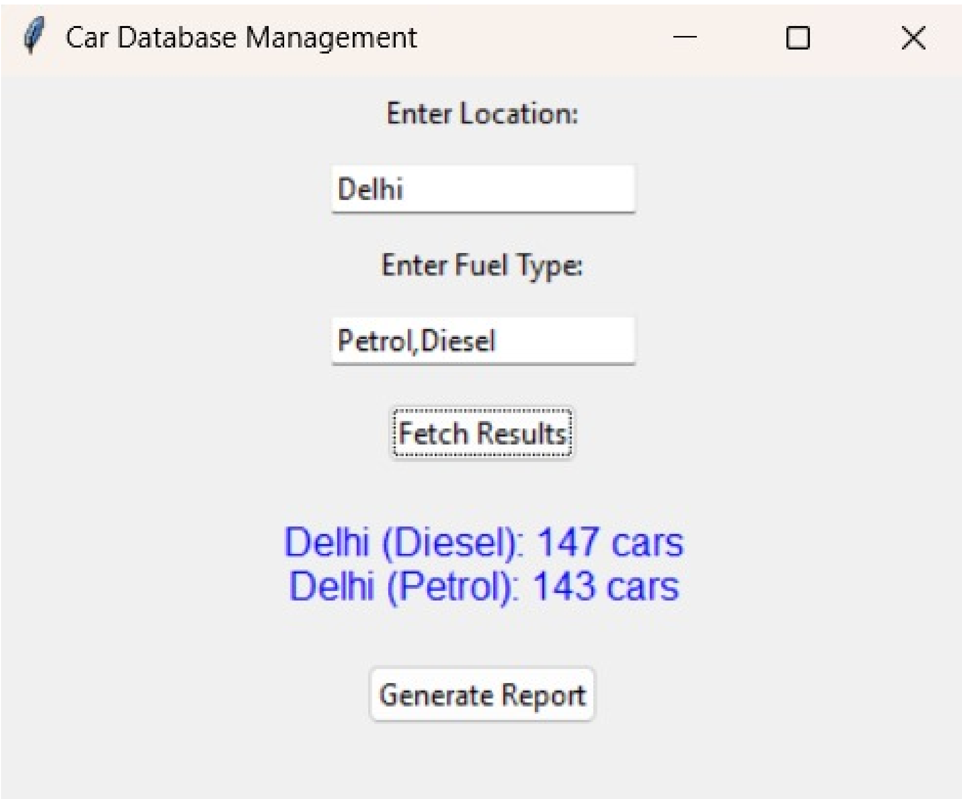

Figure 6.1: Graphical Interface Example 1

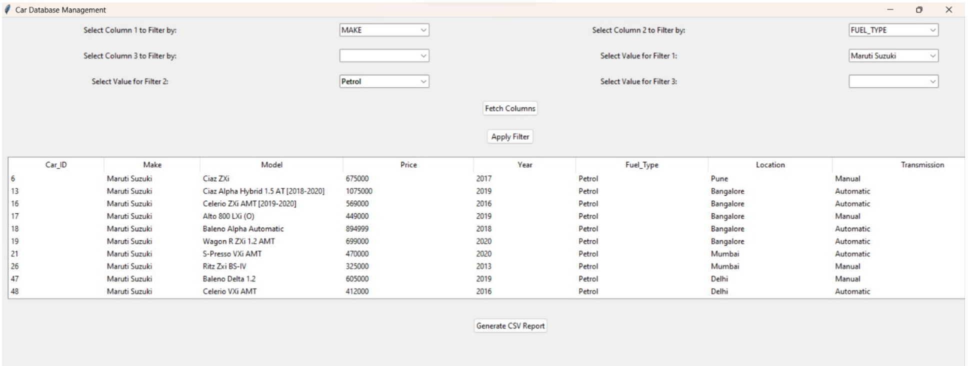

Figure 6.2: Graphical Interface Example 2

Chapter 7

Conclusion

This project effectively showcases the design, security, optimization, and automation of a car management database, enhanced by user-friendly GUIs for non-technical users. Future developments will focus on expanding machine learning capabilities for price prediction.The integration of robust data protection measures ensures that sensitive information remains secure while still being accessible to authorized personnel. Additionally, the optimization of database queries and indexing techniques significantly enhances the performance of the system, allowing for swift data retrieval and processing.

To further streamline operational efficiency, various automation features have been implemented, such as scheduled maintenance reminders and automated reporting tools, saving time and minimizing human error. The user interfaces have been meticulously designed with intuitive navigation and clear visual cues, enabling users from diverse backgrounds to interact with the database confidently.

In the upcoming phases of development, our attention will pivot towards leveraging machine learning algorithms to provide insightful analytics and predictive modeling. This exciting direction will empower users not only to forecast pricing trends but also to evaluate factors influencing car demand and market dynamics.

Overall, this project lays a solid foundation for future enhancements, positioning itself as a vital resource in the automotive industry. With continuous feedback integration and iterative improvements, we aim to cultivate an adaptive and intelligent car management system that meets the evolving needs of our users.

Scan to access our GitHub repository!